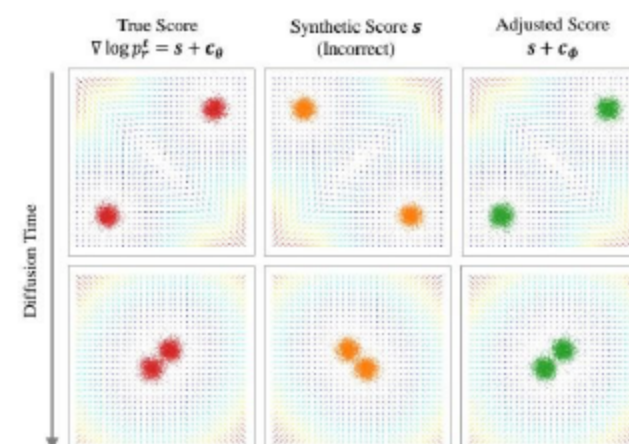


Discriminator Guidance in Diffusion Models (G++)

Dongjun Kim et al. (KAIST, NAVER Cloud, Summary.AI) • 2023

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Abstract: Key Idea and Contributions



- Discriminator Guidance refines diffusion scores
 - Supervises denoising paths without joint training
 - Stabilizes and accelerates training vs GANs
- Plug-and-play with pre-trained diffusion models
 - SOTA on ImageNet 256×256: FID 1.83 (DG)
 - Open-source code: github.com/alsdudr1a10/DG

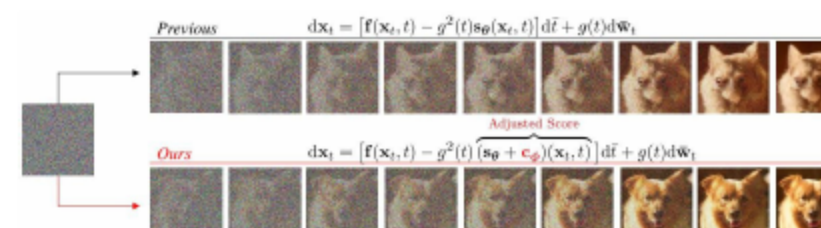
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Background: Diffusion Models and Guidance

- Diffusion models excel in image, video, and text-to-image.
- Generation relies on score estimation of data gradients.
- Guidance variants: classifier-based and classifier-free.
 - Classifier-free guidance improves quality but narrows diversity.
 - Pre-trained score models cut compute, yet bias remains.
- Motivation: add discriminators to correct score bias.

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G++: Discriminator Guidance

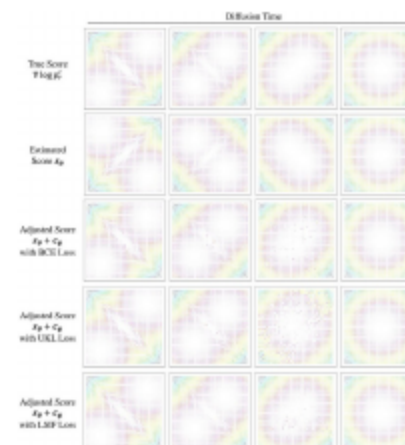


- Auxiliary discriminator guides diffusion sampling
 - Improves sample quality without retraining score models
 - Provides density-ratio signals to fix scores
- Corrects model score during each denoising step
 - Dynamic, data-dependent guidance avoids memorization
 - Plug-in module: enhance scores, keep base unchanged

5 Preliminaries – Theoretical Frameworks

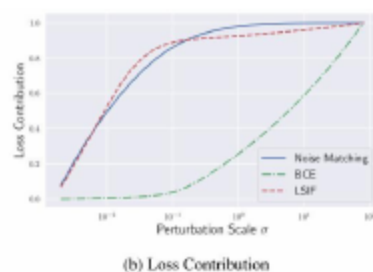
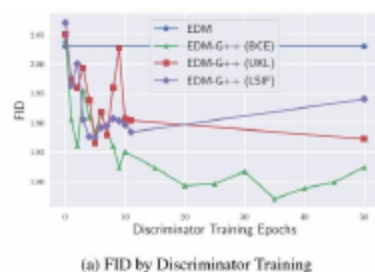
- Likelihood maximization underpins diffusion training
- Forward SDE defines the noisy path from x_0 to x_T
- Score function estimates $\text{grad}_x \log q_t(x)$ to denoise
 - Training aligns $q(x_0:T)$ with $p_\theta(x_0:T)$ via KL minimization
 - Density ratios connect NLL to ELBO and NELBO terms
- Truncated noise scales unify objectives, grounding DG

6 Discriminator-Guided Score Correction



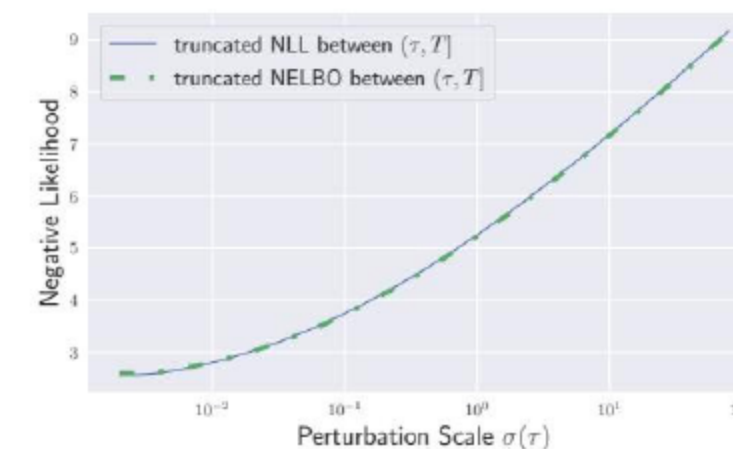
- Reverse process drifts from reality via local optima.
 - Add a discriminator-validated score correction.
 - Correction nudges model score toward true field.
- Scale guidance by noise level to avoid overstep.
 - Adjusted scores under NM, BCE, LSIF losses.
 - Improved trajectories reduce mode bias, artifacts.

7 Discriminator Training & Signal Shaping



- Classify noisy samples; estimate density ratios
 - Protocol: train on noisy-real vs gen paths
 - Losses compared: Noise-Match, BCE, LSIF; scale-aware
- FID trends across epochs; early gains
 - Temporal weighting: emphasize mid- σ ; scale-varying loss
 - Stability: avoid overconfident D; smooth loss

8 Theoretical Analysis and Optimality



- DG aligns samples with data by correcting scores
 - Guidance reduces KL vs pure model sampling
 - Optimality tied to accurate density-ratio cues
- Bias-variance trade-offs vary by noise scale
 - Lower truncated NLL emerges at proper scales
 - Score error minimization improves realism

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Experiments: FID, Diversity, I2I

X

Image Generation: Quality & Diversity



- Discriminator Guidance cuts FID vs. non-guided.
 - Improves Recall, indicating broader sample modes.
 - Consistent gains across penguin, cat, burger sets.
- CIFAR-10 and ImageNet show higher visual fidelity.
 - Quality-diversity trade-off shifts favorably.
 - Robust improvements hold under varied classes.

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Image-to-Image: Cat→Dog Trade-offs



- Cat→Dog I2I: DG boosts realism without drift.
 - PSNR-FID trade-off reveals faithfulness-realism curve.
 - SDEdit+DG attains lower FID at matched PSNR.
- Progression shows controlled Cat→Dog transitions.
 - Sweet spot balances identity faithfulness and realism.
 - Tune noise step to trade faithfulness for realism.

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Discussion, Impact, and Future Directions

- Advances state-of-the-art diffusion results across benchmarks.
- Strengthens guided sampling while improving efficiency.
- Limitations: calibration, stability, and compute costs.
 - Explore adaptive guidance schedules for robust sampling.
 - Improve ratio estimation; leverage Bregman divergences.
- Pursue joint score-discriminator training; broader modalities.